

Physiologically Guided R-Centered Deep Learning for ECG Wave Delineation Across Heterogeneous Databases

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Accurate delineation of P and T waves in electrocardiogram (ECG) signals remains challenging because of morphological variability, pathology, and differences in recording conditions across databases. This study investigates a physiologically guided CNN-BiLSTM framework based on R-peak-centered temporal representation and cross-dataset adaptation. Experiments were conducted on the QT Database (QTDB) and Lobachevsky University Electrocardiography Database (LUDB) using record-level partitions. A controlled ablation evaluated R-centered windowing, extended post-R temporal context, normalized temporal encoding, and supervised LUDB adaptation. Across three random seeds, R-centered representation improved cross-dataset performance, while extended post-R context provided a further gain. In contrast, the normalized temporal channel slightly reduced performance, showing that physiological guidance does not necessarily improve monotonically. Adaptation using 20 LUDB records improved Macro F1 to 0.8415 ± 0.0073 on 180 untouched test records. Event-level evaluation showed high P/T sensitivity but lower PPV due to false detections. Independent R-peak auditing showed high sensitivity across both databases, with more false positives on LUDB. Overall, the findings support balanced R-centered guidance rather than increasingly restrictive physiological constraints. Code and complete reproducibility materials, including exact record-level splits, implementation-level experimental parameters, and algorithmic specifications, are available at <https://github.com/vishwambharaRH/R-Centered-Deep-Learning>.

CCS Concepts: • **Computing methodologies** → **Machine learning**; • **Applied computing** → *Life and medical sciences*.

Additional Key Words and Phrases: Electrocardiography, ECG Delineation, Deep Learning, CNN-BiLSTM, R-Peak-Guided Learning, Cross-Dataset Generalization, QT Database, LUDB, Cardiac Signal Processing

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1 Introduction

1.1 Motivation and Clinical Relevance

Electrocardiography (ECG) is a widely used non-invasive method for assessing cardiac electrical activity and plays an important role in automated cardiac analysis [1]. Accurate delineation of ECG waveforms, particularly the P wave, QRS complex, and T wave, provides temporal information about cardiac depolarization and repolarization and supports subsequent clinical interpretation [1, 2]. Automated delineation can reduce manual annotation effort and enable consistent analysis of large ECG recordings. However, P and T waves are challenging to delineate because of their low amplitudes, morphological variability, and sensitivity to noise [3]. These challenges are further amplified across heterogeneous databases, where differences in signal characteristics and patient populations can affect model generalization. This motivates the use of physiological information, particularly the R peak as a temporal reference, within deep learning-based ECG delineation.

1.2 Existing Approaches and Challenges

Early ECG analysis relied on signal-processing and rule-based methods, with the Pan-Tompkins algorithm providing a widely used approach for QRS detection through filtering, differentiation, squaring, integration, and adaptive decision rules [4]. Although computationally efficient, such methods can be less robust to variations in noise and ECG morphology. Recent approaches increasingly use machine and deep learning to learn ECG representations automatically. CNNs capture local morphological patterns, while recurrent models such as LSTMs capture temporal dependencies [3, 5]. Nevertheless, P and T wave delineation remains challenging due to their lower amplitudes, morphological variability, and susceptibility to noise [2, 3]. Furthermore, differences in acquisition characteristics, patient populations, and annotation practices across databases can reduce cross-database generalization, motivating the incorporation of physiologically meaningful temporal structure.

1.3 Physiological Guidance and Cross-Dataset Generalization

Physiological information can provide useful structure for ECG delineation by exploiting the temporal relationships between cardiac waveforms. Recent studies have incorporated R-peak priors, beat-centered processing, adaptive windows, and temporal constraints to improve waveform localization [1, 6–8]. In particular, the R peak provides a natural temporal reference around which the surrounding P, QRS, and T waveforms can be represented.

However, physiological constraints can also become restrictive when ECG morphology varies across subjects or databases. This is especially relevant for cross-dataset evaluation, where differences in acquisition characteristics and waveform morphology may reduce the effectiveness of a model trained on a single database. Existing work demonstrates the potential of physiological guidance, but the contribution of individual physiological components and the appropriate degree of constraint remain less systematically evaluated [1, 5–7]. Our work therefore investigates R-centered guidance through controlled component-wise experiments while evaluating generalization from QTDB to LUDB.

1.4 Research Gap

Although deep learning has improved automated ECG delineation, existing work largely focuses on developing increasingly sophisticated architectures or incorporating physiological priors to improve waveform localization [1, 5, 7, 9]. The extent to which physiological guidance is beneficial and whether stronger constraints consistently improve generalization have been less systematically examined. This is particularly relevant for P and T waves, whose morphology varies

substantially across subjects and databases [3]. Furthermore, the contribution of individual physiological components is often difficult to isolate when multiple design changes are introduced simultaneously.

This work addresses this gap by systematically evaluating R-centered physiological guidance through controlled component-wise experiments, examining its effects on cross-dataset delineation and the balance between physiological structure and data-driven flexibility.

1.5 Contributions

The main contributions of this work are:

- A physiologically guided CNN-BiLSTM framework for P- and T-wave delineation using R-peak-centered ECG representation.
- A controlled A0–A4 ablation evaluating R-centered windows, extended post-R context, temporal encoding, and limited LUDB adaptation under consistent experimental conditions.
- Cross-dataset evaluation from QTDB to LUDB using sample-wise, event-level, boundary-level, and independent R-peak analyses, including supervised adaptation on 20 LUDB records with evaluation on 180 untouched records.

2 Related Work

2.1 Classical ECG Delineation

Early automated ECG delineation relied primarily on signal-processing and rule-based techniques, including derivative-based methods, adaptive thresholding, wavelet transforms, and morphology-based analysis [2, 4]. Pan–Tompkins established a widely used real-time QRS detection framework based on filtering, differentiation, squaring, moving-window integration, and adaptive decision rules [4]. Although such methods are computationally efficient and interpretable, their reliance on handcrafted features and predefined rules can limit robustness to noise, morphological variability, and pathological ECG patterns [2]. These limitations motivated data-driven approaches that learn ECG morphology and temporal dependencies directly from recorded signals.

2.2 Deep Learning-based ECG Delineation

Deep learning models can learn waveform morphology and temporal patterns directly from raw or minimally processed ECG signals. CNNs capture local morphological features, while recurrent architectures such as LSTMs model longer-range temporal dependencies. Accordingly, CNN-, recurrent-, and hybrid architectures have been applied to P, QRS, and T wave delineation, demonstrating improved performance over conventional approaches [3, 5, 7]. However, their performance can remain sensitive to differences in morphology, noise, and database characteristics, so strong within-database results do not necessarily translate to robust cross-database generalization.

2.3 Physiological and R-Peak-Guided Methods

Recent approaches incorporate physiological information through R-peak priors, beat-centric processing, adaptive temporal windows, and temporal constraints to improve waveform localization [1, 6–8]. The R peak provides a physiologically meaningful temporal reference for organizing surrounding P and T waves. However, the appropriate degree of physiological constraint within a learned model remains less clear, particularly when waveform morphology varies across patients or databases [1, 7]. Our work therefore examines R-centered representation and its individual refinements

through controlled ablation, rather than assuming that increasingly strong physiological guidance necessarily improves performance.

3 Methodology

3.1 Problem Formulation

Given a single-lead ECG signal $x = \{x_t\}_{t=1}^T$, the objective is to assign each sample to one of three classes: Background, P wave, or T wave. QRS-labeled samples are mapped to the Background class for the three-class delineation task. Accordingly, the reported sample-wise Macro F1 represents the average F1 across the Background, P-wave, and T-wave classes. The model therefore learns a sequence-to-sequence mapping

$$f_{\theta}(x) \rightarrow \hat{y}, \quad \hat{y}_t \in \{\text{Background}, P, T\},$$

where θ denotes the learnable model parameters. The primary objective is to obtain accurate sample-level waveform delineation while maintaining robustness when the trained model is applied to ECG recordings from a different database. In this study, physiological guidance is introduced through the R peak as a temporal reference, providing a physiologically anchored representation of the sequence. The effectiveness of this representation and its subsequent refinements is evaluated through matched ablation experiments and cross-dataset testing.

3.2 Datasets and Preprocessing

3.2.1 Datasets and Annotation Preprocessing. Two publicly available ECG databases on PhysioNet were used: the QT Database (QTDB) and the Lobachevsky University Electrocardiography Database (LUDB) [10–12]. QTDB provides waveform annotations through several annotation files; this study used the pu0 annotations for signal 0 for model development. LUDB was used for independent cross-dataset evaluation and, in the adaptation experiment, for limited supervised domain adaptation. The dataset partitions and their roles in the experiments are summarized in Table 1. QTDB comprises 105 records sampled at 250 Hz. The first recorded signal channel was used, and waveform annotations were obtained from the corresponding pu0 annotation files. LUDB contains 200 records originally sampled at 500 Hz. Its lead II signal and corresponding ii annotations were used; signals were resampled to 250 Hz and annotation locations were scaled accordingly.

For model training, annotations were converted into sample-wise labels for Background, P wave, or T wave, with QRS-labeled samples mapped to Background. Normalization parameters were derived from the QTDB training records only and subsequently applied to the QTDB validation and LUDB data.

Table 1. Datasets, record-level partitions, window counts, and sampling details

Dataset/Partition	Records	Windows	Sampling / Lead
QTDB – Training	84	87,111 (360); 87,088 (440)	250 Hz, first channel
QTDB – Validation	21	22,799 (360); 22,793 (440)	250 Hz, first channel
LUDB – Adaptation	20	194 (360); 183 (440)	500 → 250 Hz, lead II
LUDB – Held-out Test	180	1,710 (360); 1,655 (440)	500 → 250 Hz, lead II

The QTDB training set contained 64.42% Background, 12.26% P-wave, and 23.32% T-wave samples after mapping QRS-labeled samples to Background. The 360-sample windows correspond to PRE=120 and POST=240, while the 440-sample windows use PRE=120 and POST=320.

3.2.2 Record-Level Partitioning and Leakage Prevention. To prevent information leakage, all dataset partitioning was performed at the record level before window construction. For QTDB, the 105 records were divided into 84 training records and 21 validation records using a fixed random seed of 7. For LUDB, the 200 records were divided into 20 adaptation records and 180 held-out test records using a fixed random seed of 17. Thus, windows from the same record could not appear across different partitions.

3.2.3 Preprocessing and Window Construction. Both databases were processed at 250 Hz. QTDB was used at its native sampling rate, while LUDB was resampled from 500 Hz using polyphase resampling (up=1, down=2), with annotation locations scaled accordingly. Signals were normalized using statistics derived from the QTDB training data. Input windows were either 360 samples (120 pre-R, 240 post-R) or 440 samples (120 pre-R, 320 post-R), depending on the experimental configuration. R-centered windows aligned the detected R peak to a fixed temporal position, while the baseline used fixed non-R-centered 360-sample windows with a stride of 256 samples. For sample-wise evaluation, predictions from the generated windows were pooled across all evaluated windows and compared with the corresponding window labels; continuous full-record predictions were not reconstructed.

3.3 CNN-BiLSTM Architecture

The proposed delineation model combines a convolutional neural network (CNN) with a bidirectional long short-term memory (BiLSTM) network to capture complementary morphological and temporal characteristics of ECG signals. The CNN extracts local morphological features from the input waveform, while the BiLSTM models temporal dependencies in both forward and backward directions. CNN-recurrent architectures have been widely explored for ECG segmentation and waveform delineation [3, 5, 7]. The CNN feature extractor consists of two one-dimensional convolutional blocks. The first uses 32 filters with kernel size 7, followed by batch normalization and ReLU activation, while the second uses 64 filters with kernel size 5, again followed by batch normalization and ReLU. The resulting feature sequence is passed to a single-layer bidirectional LSTM with 128 hidden units per direction. A dropout rate of 0.3 is applied before a linear layer maps the resulting 256-dimensional representation at each time step to the waveform classes. The complete CNN-BiLSTM architecture is shown in Fig. 1 and serves as the baseline throughout the study.

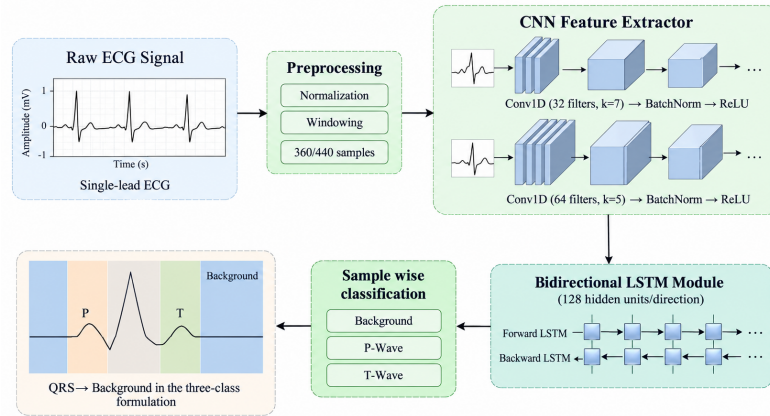


Fig. 1. Proposed CNN-BiLSTM baseline architecture for ECG wave delineation.

3.4 R-Peak Detection and R-Centered Representation

R peaks were detected using the Pan–Tompkins algorithm, providing a temporal reference for ECG delineation [4]. The detected peaks were used to construct beat-centered input windows, with the R peak positioned at a fixed reference location. As illustrated in Fig. 2, this representation provides a consistent temporal anchor for surrounding waveforms. For the primary R-centered configuration, 120 samples preceding and 240 samples following the R peak were used, giving a 360-sample window. An extended configuration increased the post-R region to 320 samples, producing a 440-sample window. The R-centered representation was evaluated against a fixed, non-R-centered window in the matched ablation study.

An independent R-peak audit was also performed on QTDB and LUDB. Detected peaks were matched one-to-one with reference annotations using a 100 ms tolerance, enabling sensitivity, positive predictive value, F1 score, and timing error to be assessed independently of downstream delineation.

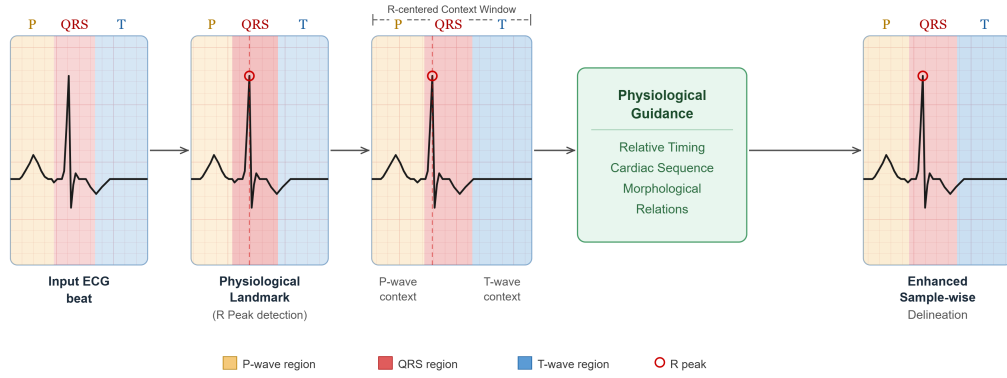


Fig. 2. Proposed R-guided framework for ECG wave delineation.

3.5 Training Procedure

The CNN–BiLSTM models were trained on the QTDB training records with performance monitored on the held-out QTDB validation set. For the matched A0–A4 experiments, Adam was used with a learning rate of 1×10^{-3} and batch size 64. R1 used cross-entropy, R2 used weighted Focal Loss with class weights 0.5175, 2.7187, and 1.4292 for Background, P, and T, respectively, with $\gamma = 2.0$, while R3 and A0–A4 used unweighted Focal Loss with $\gamma = 2.0$. No early stopping was applied. The R1–R6 results are historical developmental runs. Retrained configurations selected the checkpoint with the lowest QTDB validation loss; R4 used a frozen R1 checkpoint, while R6 used a fixed pretraining checkpoint followed by 8 epochs of adaptation. These runs therefore do not follow the fully controlled A0–A4 protocol. For the matched ablation, A0–A3 were trained for 15 epochs with seeds 1, 2, and 3, selecting the checkpoint with the highest QTDB validation Macro F1. A4 initialized from the corresponding best A3 checkpoint for each seed and underwent 8 epochs of supervised adaptation on 20 LUDB records. The final adaptation epoch was evaluated on the remaining 180 untouched LUDB records.

3.6 Experimental Configurations

3.6.1 R1–R6 Developmental Experiments. The initial development proceeded through six sequential configurations, R1–R6. R1 used standard cross-entropy loss, while R2 introduced weighted focal loss to address class imbalance. Because

weighting increased minority-class recall but reduced precision and overall performance, R3 adopted unweighted focal loss and was selected as the reference baseline. The subsequent configurations investigated physiological guidance. R4 applied an R-aware hard decoder to frozen R1 predictions and was treated as a failure-analysis experiment rather than a controlled retraining comparison. R5 introduced R-peak auditing, R-centered representation, and extended post-R context (POST320), while R6 additionally incorporated a normalized temporal-position channel and supervised adaptation using 20 LUDB records. Because multiple components changed across parts of the R1–R6 progression, these experiments are presented as developmental evidence, while component-level conclusions are based primarily on the matched A0–A4 ablation.

3.6.2 A0–A4 Matched Ablation. To evaluate the incremental effects of the proposed components, a matched A0–A4 study was performed using consistent record-level splits, preprocessing, evaluation protocol, and random seeds. A0 used fixed non-R-centered 360-sample windows. A1 introduced R-centered windows with PRE=120 and POST=240, A2 extended the post-R context to POST320, and A3 added the normalized temporal-position channel.

A4 used the corresponding best A3 checkpoint for each seed as initialization, followed by 8 epochs of supervised adaptation using the 20-record LUDB adaptation set. The remaining 180 LUDB records were kept untouched for final evaluation. A0–A3 were trained for 15 epochs with seeds 1, 2, and 3.

4 Experimental Results

4.1 Matched Component Ablation: A0–A4

The matched ablation evaluated the incremental effects of the proposed physiological guidance components under the common protocol described in Section 3.6.2, with results summarized in Table 2. All variants were evaluated on the same 180 untouched LUDB test records. R-centered window construction increased Macro F1 from 0.7999 to 0.8202, while extending post-R context to POST320 further improved performance to 0.8308. Adding the normalized temporal-position channel reduced Macro F1 slightly to 0.8263. Finally, supervised adaptation using 20 LUDB records increased Macro F1 to 0.8415 ± 0.0073 on the 180-record held-out test set. Overall, the ablation indicates that R-centered window construction and extended post-R context were beneficial, whereas additional temporal encoding was not; limited target-domain adaptation provided a further improvement.

Table 2. Matched A0–A4 ablation results on the LUDB test set. Values show Macro F1 as mean $\pm$ standard deviation across three random seeds.

Configuration	Description	LUDB Macro F1
A0	Fixed, non-R-centered baseline	0.7999 $\pm$ 0.0042
A1	R-centered representation	0.8202 $\pm$ 0.0009
A2	A1 + extended post-R context (POST320)	0.8308 $\pm$ 0.0078
A3	A2 + normalized temporal-position channel	0.8263 $\pm$ 0.0061
A4	A3 + supervised LUDB adaptation	0.8415 $\pm$ 0.0073

4.2 Baseline Development and Failure Analysis: R1–R3

Three baseline training strategies were evaluated using the same CNN–BiLSTM architecture to establish a reference model and assess class-imbalance handling. Results are summarized in Table 3. R1 used standard cross-entropy loss and achieved Macro F1 scores of 0.9026 on QTDB and 0.8203 on LUDB. R2 introduced weighted focal loss, increasing P- and T-wave recall but reducing precision and overall performance; Macro F1 decreased to 0.8420 and 0.7604, respectively. R3

used unweighted focal loss and restored the precision–recall balance, achieving the strongest baseline performance with Macro F1 scores of 0.9061 on QTDB and 0.8236 on LUDB. A representative QTDB validation ECG segment showing the ground-truth annotations and R3 predictions is presented in Fig. 3. R3 was therefore selected as the reference baseline for subsequent physiological-guidance experiments.

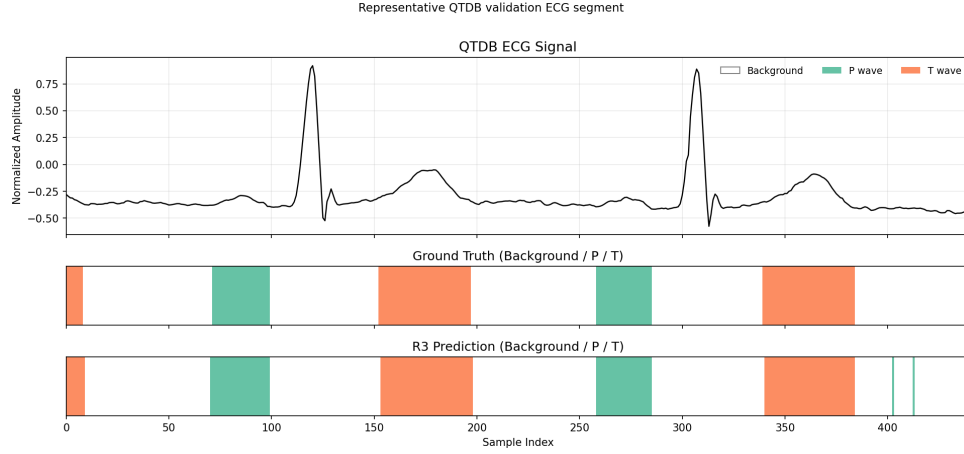


Fig. 3. Representative QTDB validation ECG segment (index 15) used for qualitative visualization.

4.3 Framework Development: R4–R6

Following selection of R3 as the reference baseline, R4–R6 explored physiological guidance and cross-dataset adaptation as developmental configurations rather than controlled component-wise comparisons. Results are summarized in Table 3. R4 applied a frozen R-aware decoder to R1 predictions and reduced Macro F1 to 0.7328 on QTDB and 0.7119 on LUDB, suggesting that rigid physiological decoding can overly restrict valid predictions. R5 introduced R-centered representation and POST320 context, restoring Macro F1 to 0.9071 and 0.8266, respectively. R6 additionally introduced the temporal-position channel and supervised adaptation using 20 LUDB records, achieving 0.9122 on QTDB and 0.8451 on LUDB. A representative LUDB held-out test segment comparing R3 and R6 is shown in Fig. 4. The R3 and R6 LUDB results use different evaluation protocols: R3 represents direct QTDB-to-LUDB generalization, whereas R6 uses adaptation followed by evaluation on 180 held-out records; their numerical difference therefore does not provide a controlled estimate of the adaptation effect. These developmental results motivated the matched A0–A4 ablation.

Table 3. Developmental R1–R6 configurations and Macro F1 performance on QTDB and LUDB.

Experiment	Configuration / Main Change	QTDB Macro F1	LUDB Macro F1
R1	Cross-Entropy baseline	0.9026	0.8203
R2	Weighted Focal Loss	0.8420	0.7604
R3	Unweighted Focal Loss	0.9061	0.8236
R4	Frozen R-aware Decoder	0.7328	0.7119
R5	R-centered + POST320	0.9071	0.8266
R6	Temporal Channel + LUDB Adaptation	0.9122	0.8451

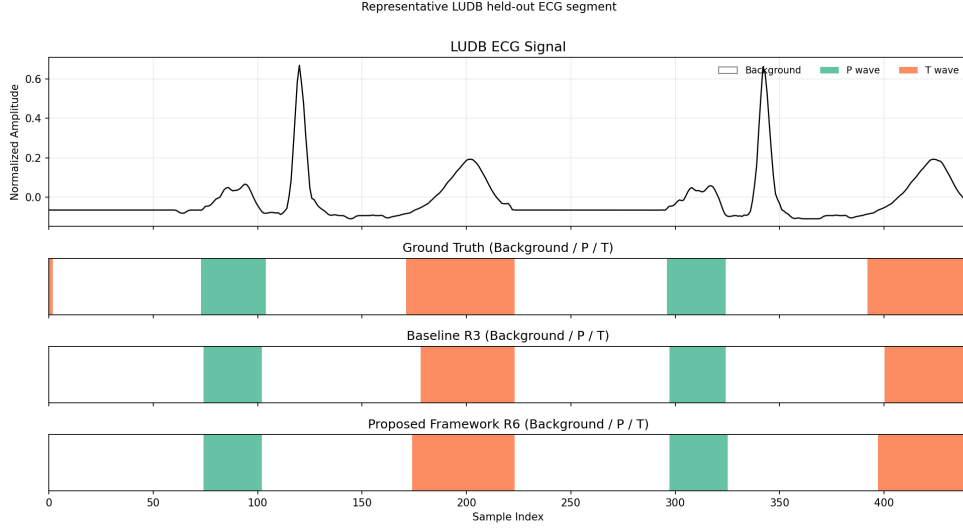


Fig. 4. Representative LUDB held-out test ECG segment (index 20) used for qualitative comparison.

4.4 Event-Level Delineation Evaluation

Because sample-wise Macro F1 does not directly assess individual event detection, event-level evaluation was performed on the LUDB test set for the historical R6 configuration using one-to-one matching. These results are supplementary to the primary controlled A0–A4 ablation. A predicted P- or T-wave event was considered a match when it occurred within a predefined tolerance of 25 samples (100 ms at 250 Hz) of the corresponding reference event. Each prediction and reference event could participate in at most one match; unmatched predictions and reference events were counted as false positives and false negatives, respectively. Across the three random seeds, sensitivity remained high for both waveform types, while PPV was substantially lower. P-wave sensitivity ranged from 0.9538–0.9812 and PPV from 0.6352–0.6575, while T-wave sensitivity ranged from 0.9627–0.9709 and PPV from 0.6316–0.6530. Event-level F1 ranged from 0.7625–0.7869 for P waves and 0.7653–0.7782 for T waves. The seed-wise results are reported in Table 4.

Table 4. Event-level P- and T-wave delineation performance across three random seeds on the LUDB test set.

Seed	P Sensitivity	P PPV	P F1	T Sensitivity	T PPV	T F1
1	0.9796	0.6575	0.7869	0.9627	0.6530	0.7782
2	0.9812	0.6464	0.7793	0.9709	0.6316	0.7653
3	0.9538	0.6352	0.7625	0.9668	0.6486	0.7764

4.5 Boundary-Level Delineation

To further assess temporal delineation accuracy, boundary-level errors were evaluated for P and T waves on the 180-record LUDB test set using the matched events described in Section 4.4. Predicted onset and offset boundaries were compared with the corresponding reference boundaries, and errors were reported in milliseconds. Positive values indicate later and negative values earlier predictions. Results across the three random seeds are summarized in Table 5. P-wave boundary errors remained relatively low across seeds, with onset and offset MAE generally around 10–13 ms.

T-wave onset showed the largest boundary error, with MAE around 24–26 ms, while T-wave offset errors were lower at approximately 16–18 ms. Signed errors indicated a tendency toward later onset and earlier offset predictions. Boundary measurements were computed only for successfully matched events and therefore complement event-level sensitivity and PPV.

Table 5. P- and T-wave boundary delineation MAE across three random seeds on the LUDB test set.

Wave	Boundary	Seed 1 MAE (ms)	Seed 2 MAE (ms)	Seed 3 MAE (ms)
P	Onset	12.43	11.27	10.02
P	Offset	11.08	12.79	12.69
T	Onset	23.67	25.61	23.79
T	Offset	16.11	17.21	18.31

4.6 Independent R-Peak Audit

Because the proposed delineation framework uses the R peak as its physiological temporal anchor, the underlying R-peak detector was evaluated independently from the downstream P/T delineation model. The deterministic detector was assessed separately on QTDB and LUDB using one-to-one matching with a 100-ms tolerance. Results are summarized in Table 6. On QTDB, using the first recorded signal channel and the `pu0` annotations, the detector achieved a mean record-level sensitivity of 0.9780, PPV of 0.9863, and F1-score of 0.9804, with a mean timing error of 13.31 ms. On LUDB, using lead II after resampling from 500 Hz to 250 Hz and scaling the annotations accordingly, sensitivity was 0.9753, PPV 0.7971, and F1-score 0.8800, with a mean timing error of 3.07 ms.

Table 6. Independent R-peak detection performance on QTDB and LUDB.

Dataset	Sensitivity	PPV	F1	Mean Timing Error (ms)
QTDB	0.9780	0.9863	0.9804	13.31
LUDB	0.9753	0.7971	0.8800	3.07

4.7 R-Localization Errors and Downstream Delineation

Because the proposed framework relies on R-peak localization as its temporal reference, we examined associations between R-detector errors and downstream P- and T-wave delineation on the same 180 untouched LUDB test records. R-error measures included total errors, missed-peak rate, false-positive rate, and timing error, while downstream performance was measured using P-wave F1, T-wave F1, and P/T Macro F1. Spearman correlations are summarized in Table 7. Total R errors showed significant negative associations with P-wave F1 ($\rho = -0.3254$, $p < 0.001$), T-wave F1 ($\rho = -0.2293$, $p = 0.0020$), and P/T Macro F1 ($\rho = -0.3342$, $p < 0.001$). Because total error counts can depend on the number of beats or record duration, missed-R and false-positive rates provide normalized measures and are more directly interpretable across records. Missed-R rate showed a similar pattern, while false-positive R rate was significantly associated with P-wave F1 and P/T Macro F1 but not T-wave F1. R-peak timing error showed weaker associations, with only the P/T Macro F1 correlation reaching statistical significance ($\rho = -0.1660$, $p = 0.0263$). Overall, greater R-detection error rates were associated with lower downstream delineation performance, particularly for P waves. These observational associations do not establish causality, and the results indicate that R-localization errors alone do not fully explain downstream performance.

Table 7. Spearman correlations between R-localization error measures and downstream P- and T-wave F1 scores on the LUDB test set.

R-error measure	P F1 Spearman ρ	T F1 Spearman ρ	P/T Macro F1 Spearman ρ
Total R errors	−0.3254**	−0.2293**	−0.3342***
Missed R rate	−0.3191**	−0.1495*	−0.3043***
False-positive R rate	−0.3149**	−0.1035	−0.1752*
R timing error	−0.1170	−0.0784	−0.1660*

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

4.8 Qualitative and Error Analysis

Record-level analysis revealed substantial variation across LUDB recordings, particularly in cases with poor R-peak localization. Three difficult records (90, 93, and 74) exhibited P-wave F1 of 0 together with severe R-detection errors. Record 90 had 18 total R errors, while Records 93 and 74 had 16 and 15, respectively. Despite the severe P-wave degradation, T-wave F1 remained comparatively higher at 0.7615, 0.7770, and 0.8849, respectively. These cases complement the correlation analysis in Section 4.7 and illustrate that poor R localization can coincide with severe P-wave degradation, while R-localization errors alone do not fully determine T-wave performance. Reliable record-level pathology labels were not available in the current evaluation pipeline; therefore, no pathology-specific interpretation is made.

5 Discussion

5.1 Interpretation of the Main Findings

The matched ablation indicates that R-centered representation and extended post-R context improve cross-dataset ECG delineation, whereas the normalized temporal-position channel provides no additional benefit. This non-monotonic trend suggests that increasing physiological information does not necessarily improve performance. The R4 failure further suggests that rigid physiological constraints can overly restrict model predictions. Limited adaptation using 20 LUDB records subsequently improved performance to 0.8415 ± 0.0073 on 180 untouched records. R-peak detector auditing showed high sensitivity but substantially lower PPV on LUDB. Overall, the findings support moderate R-centered physiological guidance as a balance between physiological structure and learned flexibility.

5.2 Comparison, Limitations, and Future Directions

The framework shows encouraging cross-dataset performance, but direct superiority over existing methods cannot be claimed because no matched external comparator was evaluated. Limitations include the QTDB-to-LUDB setting, single-lead evaluation, limited 20-record adaptation, higher R-peak false-positive rate on LUDB, unavailable pathology-specific analysis, and the absence of QRS boundary delineation in the three-class formulation and the use of a single QTDB record-level split, with three seeds used to assess training variability. Future work should include multi-database and multi-lead validation, improved R-peak robustness, pathology-stratified analysis, and matched comparisons with established delineation methods.

6 Conclusion

This study investigated a physiologically guided CNN-BiLSTM framework for ECG waveform delineation across heterogeneous databases. The matched ablation demonstrated that R-centered representation and extended post-R temporal context improved cross-dataset performance, while additional temporal encoding did not provide further benefit.

Limited adaptation using 20 LUDB records further improved performance on 180 untouched test records, achieving a Macro F1 of 0.8415 ± 0.0073 . Independent R-peak analysis showed generally reliable localization while highlighting increased false-positive detections on LUDB. Overall, the findings support moderate R-centered physiological guidance combined with learned temporal representations and limited domain adaptation as a promising strategy for robust cross-dataset ECG delineation.

7 Ethics and Disclosures

LLMs have been used for grammar validation and spell checking, and generated content has been validated.

References

- [1] Jilong Wang, Renfa Li, Rui Li, and Bin Fu. A knowledge-based deep learning method for ecg signal delineation. *Future Generation Computer Systems*, 109:56–66, 2020. ISSN 0167-739X. doi: <https://doi.org/10.1016/j.future.2020.02.068>.
- [2] Nourhan Bayasi, Temesghen Tekeste, Hani Saleh, Ahsan Khandoker, Baker Mohammad, and Mohammed Ismail. Adaptive technique for p and t wave delineation in electrocardiogram signals. In *2014 36th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, pages 90–93, 2014. doi: <https://doi.org/10.1109/EMBC.2014.6943536>.
- [3] Annisa Darmawahyuni, Siti Nurmaini, Muhammad Naufal Rachmatullah, Prazna Paramitha Avi, Samuel Benedict Putra Teguh, Ade Iriani Sapitri, Bambang Tutuko, and Firdaus Firdaus. Improved delineation model of a standard 12-lead electrocardiogram based on a deep learning algorithm. *BMC Medical Informatics and Decision Making*, 23(1):139, Jul. 2023. ISSN 1472-6947. doi: <https://doi.org/10.1186/s12911-023-02233-0>.
- [4] Jiapu Pan and Willis J. Tompkins. A real-time qrs detection algorithm. *IEEE Transactions on Biomedical Engineering*, BME-32(3):230–236, 1985. doi: <https://doi.org/10.1109/TBME.1985.325532>.
- [5] Xiaohong Liang, Liping Li, Yuanyuan Liu, Dan Chen, Xinpei Wang, Shunbo Hu, Jikuo Wang, Huan Zhang, Chengfa Sun, and Changchun Liu. Ecg_segnet: An ecg delineation model based on the encoder-decoder structure. *Computers in Biology and Medicine*, 145:105445, 2022. ISSN 0010-4825. doi: <https://doi.org/10.1016/j.compbiomed.2022.105445>.
- [6] Zhenqin Chen, Yiwei Lin, Yuying Bao, Fengdan Hu, and Jinshan Xu. Glf-net: A quasi-periodic prior-guided waveform segmentation network for ecg delineation. *Artificial Intelligence in Medicine*, 180:103485, 2026. ISSN 0933-3657. doi: <https://doi.org/10.1016/j.artmed.2026.103485>.
- [7] Mario Sanz-Guerrero, Sergio González-Cabeza, Luis Piñuel, Mauro Luis Buelga Suárez, Gonzalo Luis Alonso Salinas, Marian Diaz-Vicente, and Joaquín Recas. Peak attention u-net: Enhancing ecg delineation with attention. *Biomedical Signal Processing and Control*, 119:109874, 2026. ISSN 1746-8094. doi: <https://doi.org/10.1016/j.bspc.2026.109874>.
- [8] Yutaka Yoshida and Kiyoko Yokoyama. Sample-wise false-positive reduction in ecg p-, r-, and t-peak detection via physiological temporal constraints and lightweight binary classifiers. *Signals*, 7(2):28, 2026. ISSN 2624-6120. doi: <https://doi.org/10.3390/signals7020028>.
- [9] Chankyu Joung, Mijin Kim, Taejin Paik, Seong-Ho Kong, Seung-Young Oh, Won Kyeong Jeon, Jae-Hu Jeon, Joong-Sik Hong, Wan-Joong Kim, Woong Kook, Myung-Jin Cha, and Otto van Koert. Deep learning based ECG segmentation for delineation of diverse arrhythmias. Apr. 2024. doi: <https://doi.org/10.1371/journal.pone.0303178>.
- [10] Pablo Laguna, Roger G Mark, Ary L Goldberger, and George B Moody. A database for evaluation of algorithms for measurement of qt and other waveform intervals in the ecg. In *Computers in Cardiology 1997*, volume 24, pages 673–676. IEEE, 1999.
- [11] Alena Kalyakulina, Igor Yusipov, Viktor Moskalenko, Alexander Nikolskiy, Konstantin Kosonogov, Nikolai Zolotykh, and Mikhail Ivanchenko. Lobachevsky University Electrocardiography Database. *PhysioNet*, Jan. 2021. doi: <https://doi.org/10.13026/eeegm-h675>. Version 1.0.1.
- [12] Tom Pollard, Benjamin E. Moody, Li wei Lehman, Brian Gow, Chrystinne Fernandes, Chen Xie, Alistair Johnson, Roger G. Mark, and Thomas Heldt. Physionet as a global platform for biomedical research. *Nature Health*, 1(8):792–795, 2026. ISSN 3005-0693. doi: <https://doi.org/10.1038/s44360-026-00096-z>.

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